



KLE Technological University

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HydroOps: AI-Driven Smart Water Pressure Monitoring and Management System

Course Name: Embedded Systems and IoT

Course Code: 25ECSC308

Semester: VI (CSE)

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May 2026

ACKNOWLEDGEMENT

We express our sincere gratitude to our guide, Dr. Vijayalakshmi M, Head of the Department, School of Computer Science and Engineering, KLE Technological University, Hubballi, for her valuable guidance, constant encouragement, and insightful suggestions throughout the development of this project. Her support and motivation played a vital role in the successful completion of this work.

We would also like to extend our heartfelt thanks to Ms. Geeta Sannakki and Ms. Neha Panchal for their continuous support, constructive feedback, and encouragement during various stages of the project.

We express our gratitude to Dr. Narayan D. G., Head of the School of Computer Science and Artificial Intelligence, for his valuable support and motivation. We also thank Dr. B. S. Anami, Registrar, Dr. Prakash Tewari, Pro Vice-Chancellor, and Dr. Ashok S. Shettar, Vice-Chancellor, KLE Technological University, for providing excellent academic facilities and a conducive environment for learning and innovation.

We are grateful to KLE Technological University for providing the infrastructure, laboratory facilities, computational resources, and opportunities that enabled us to successfully complete this project.

We also acknowledge the support received from the organizers of the SAMVED Smart Governance Hackathon, MIT Vishwaprayag University, and Solapur Municipal Corporation, whose problem statement and expert feedback helped us validate and refine our proposed solution.

Finally, we would like to thank all the teaching and non-teaching staff members of the School of Computer Science and Engineering, our friends, and our well-wishers for their encouragement, cooperation, and support throughout the project.

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ABSTRACT

Efficient water distribution remains a significant challenge for urban local bodies due to issues such as leakage, unequal supply, delayed fault detection, and increasing operational complexity. Traditional monitoring approaches rely heavily on manual supervision and often lack real-time visibility into the condition of the distribution network. To address these challenges, this project proposes *HydroOps*, an AIoT-based Smart Water Pressure Monitoring and Management System designed for intelligent municipal water distribution.

The proposed system integrates IoT sensing devices, embedded controllers, cloud-based MQTT communication, machine learning analytics, and an Android application into a unified platform. Flow sensors and ultrasonic sensors continuously monitor water flow and Elevated Service Reservoir (ESR) levels. Sensor data is transmitted through ESP8266 communication modules to a cloud-connected MQTT broker, where analytical services process the incoming telemetry in real time.

HydroOps incorporates intelligent features such as leak and burst detection using mass-balance analysis, anomaly detection using machine learning models, demand forecasting based on population and LPCD calculations, automated valve control, and citizen feedback integration. The system also provides AI-driven alerts and recommendations to assist municipal engineers in operational decision-making. A DBSCAN-based clustering mechanism is utilized to identify complaint hotspots from citizen reports, enabling faster identification of recurring issues.

The developed prototype demonstrates real-time monitoring, intelligent analytics, automated response mechanisms, and scalable cloud connectivity. The results indicate that HydroOps can significantly improve operational efficiency, reduce water losses, support equitable water distribution, and enhance citizen participation in municipal water management.

Keywords: *Artificial Intelligence of Things (AIoT), Smart Water Management, MQTT, IoT, Leak Detection, Demand Forecasting, Machine Learning, Citizen Feedback System, Municipal Water Distribution*

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Chapter 1

INTRODUCTION

Water distribution is one of the most critical services managed by urban municipal corporations. Rapid urbanization, increasing population density, aging pipeline infrastructure, and inefficient monitoring systems have significantly increased the complexity of municipal water management. In many cities, water supply networks experience frequent issues such as pressure imbalance, leakage, unauthorized consumption, irregular supply scheduling, and delayed fault detection. These problems result in non-revenue water losses, unequal water distribution, and operational inefficiencies.

Traditional water monitoring systems primarily depend on manual inspection and reactive maintenance approaches. Faults are often identified only after visible infrastructure damage or citizen complaints are reported. The absence of real-time telemetry and intelligent analytics limits the ability of engineers to make proactive operational decisions. Therefore, there is a growing need for smart water management systems capable of continuous monitoring, automated analysis, and intelligent decision support.

Recent advancements in Internet of Things (IoT), cloud communication, artificial intelligence, and embedded systems have enabled the development of intelligent infrastructure management solutions. By integrating sensors, communication modules, machine learning algorithms, and mobile applications, water distribution systems can be transformed into smart and responsive networks.

HydroOps is developed as an AIoT-based smart water pressure monitoring and management system for municipal water distribution networks. The system combines real-time sensing, MQTT-based cloud communication, machine learning analytics, and intelligent automation into a unified platform. The proposed system is capable of monitoring ESR levels, detecting leakages, identifying abnormal flow patterns, forecasting water demand, supporting automated valve control, and integrating citizen feedback mechanisms.

1.1 Motivation

Urban local bodies frequently face challenges in ensuring equitable and efficient water supply across all distribution zones. Tail-end and elevated regions often receive inadequate supply due to pressure drops and uneven scheduling. In addition, pipeline leakages and unauthorized consumption contribute significantly to water losses and operational costs.

Most existing systems focus only on displaying sensor data without providing intelligent

insights or automated assistance. Municipal operators are therefore required to manually analyze conditions and make decisions under time-sensitive situations. This motivated the development of HydroOps as a system capable of not only monitoring the network but also analyzing live telemetry data and generating actionable recommendations.

Another important motivation behind this work was to bridge the gap between conventional SCADA-style monitoring systems and modern AI-driven infrastructure management. The project also aims to encourage smart city development by integrating citizen participation, intelligent analytics, and real-time automation into municipal operations.

1.2 Problem Statement

Solapur Municipal Corporation faces several challenges in managing urban water distribution effectively. Uneven pressure distribution frequently affects tail-end and elevated areas, leading to irregular water supply and citizen dissatisfaction. Aging pipelines and delayed maintenance further increase the occurrence of leakages and burst conditions, resulting in large quantities of water loss.

Manual valve operations and the absence of centralized monitoring systems reduce operational efficiency and increase response time during emergencies. Furthermore, the lack of analytical tools for demand forecasting and anomaly detection limits the ability of engineers to optimize water distribution schedules.

Hence, there is a requirement for an intelligent system capable of:

- Monitoring water flow and ESR levels continuously
- Detecting leakages and abnormal conditions in real time
- Forecasting demand dynamically
- Supporting automated valve control
- Assisting operators using intelligent alerts and recommendations
- Integrating citizen complaints into the operational workflow

1.3 Objectives

The primary objective of HydroOps is to design and develop an AI-driven smart water pressure monitoring and management system capable of improving operational efficiency and ensuring equitable water distribution.

The specific objectives of the project are as follows:

1. To continuously monitor water flow and ESR levels using IoT-based sensing devices.
2. To establish real-time communication between field devices and cloud infrastructure using MQTT protocol.
3. To detect leakages, abnormal flow conditions, and unauthorized consumption using analytical and machine learning techniques.
4. To forecast water demand using ESR availability, population statistics, and LPCD-based scheduling methods.
5. To provide intelligent recommendations and emergency response mechanisms for municipal engineers.
6. To develop an Android-based application for monitoring, control, alerts, and citizen feedback integration.
7. To design a scalable architecture suitable for future smart city deployment.

1.4 Scope of the Project

HydroOps is implemented as a prototype-scale smart water distribution system integrating IoT hardware, cloud communication, machine learning analytics, and mobile applications. The system demonstrates how real-time sensing and intelligent automation can improve municipal water management operations.

The project includes:

- ESR water level monitoring using ultrasonic sensors
- Flow monitoring using multiple flowmeters
- MQTT-based telemetry communication
- Leak and anomaly detection mechanisms

- Demand forecasting and water scheduling
- Valve control and automation
- Android-based operator dashboard
- Citizen complaint and feedback integration

Although the current implementation uses a scaled physical prototype, the software architecture is designed to support future city-scale deployment.

1.5 Organization of the Report

The remainder of the report is organized as follows:

- Chapter 2 presents the literature survey and related work in smart water management systems.
- Chapter 3 explains the overall system architecture and design methodology of HydroOps.
- Chapter 4 describes the hardware implementation, sensing layer, and communication workflow.
- Chapter 5 discusses the analytical models, machine learning modules, and demand forecasting techniques used in the system.
- Chapter 6 presents the implementation details, experimental results, and performance analysis.
- Chapter 7 concludes the report and discusses future enhancements and deployment possibilities.

Chapter 2

LITERATURE SURVEY

The rapid growth of urban populations and increasing pressure on municipal infrastructure have created a strong need for intelligent water management systems. Traditional water distribution systems primarily depend on manual supervision and delayed maintenance approaches, which often result in excessive water losses, unequal distribution, and delayed fault detection. Recent advancements in Internet of Things (IoT), cloud computing, artificial intelligence, and embedded systems have enabled the development of smart water monitoring and management solutions capable of real-time sensing, automated analysis, and intelligent decision support.

Several researchers have proposed IoT-based architectures for monitoring water flow, tank levels, and pipeline conditions. These systems commonly use embedded controllers, flow sensors, and wireless communication technologies to transmit telemetry data to cloud platforms. MQTT has emerged as one of the most widely adopted lightweight communication protocols for real-time IoT telemetry due to its low bandwidth requirements and publish-subscribe architecture.

Many existing smart water systems focus primarily on monitoring and visualization. Although these systems provide real-time dashboards and basic alerts, they often lack advanced analytical capabilities such as predictive demand forecasting, intelligent fault prioritization, automated control mechanisms, and citizen feedback integration.

Machine learning techniques have recently gained importance in infrastructure monitoring applications. Algorithms such as Isolation Forest, clustering techniques, and anomaly detection models are increasingly used to identify unusual patterns in sensor data. These methods help improve fault detection accuracy and reduce response time during operational emergencies.

Several studies have also explored the use of smart valves and automated control systems in municipal water networks. Automated actuation allows rapid response during critical leak or burst conditions, thereby reducing water losses and infrastructure damage. However, many existing implementations are limited to simulation environments and are not integrated with real-time mobile applications or cloud-based analytics platforms.

In addition to operational monitoring, citizen participation has become an important component of modern smart city systems. Complaint-driven maintenance systems and geolocation-based issue reporting platforms help municipal authorities identify service disruptions and improve operational transparency. However, the integration of citizen feed-

back with real-time telemetry and AI-driven analytics remains limited in many existing solutions.

HydroOps attempts to address these limitations by combining multiple technologies into a unified smart water management platform. The proposed system integrates:

- IoT-based sensing for real-time telemetry acquisition
- MQTT-based cloud communication for low-latency data exchange
- Machine learning techniques for anomaly and leak detection
- Demand forecasting using ESR availability and LPCD calculations
- Automated valve control mechanisms
- Android-based dashboards for engineers and citizens
- Citizen complaint integration and feedback analysis

Unlike conventional monitoring systems that focus only on visualization, HydroOps emphasizes intelligent decision support and operational automation. The system continuously analyzes telemetry data, prioritizes alerts, generates AI-driven recommendations, and supports emergency response actions during critical conditions.

2.1 Existing Smart Water Monitoring Systems

Several smart water monitoring systems have been developed using IoT technologies to improve visibility into municipal water distribution networks. Most of these systems use flow sensors, water level sensors, and embedded microcontrollers for collecting real-time telemetry data.

Cloud-based dashboards are commonly used to visualize sensor readings and monitor operational parameters remotely. Some systems also integrate GSM, Wi-Fi, or LoRa communication technologies for transmitting sensor data over long distances.

Although these systems improve monitoring capabilities, many of them are limited to basic telemetry visualization and threshold-based alert generation. Advanced features such as automated decision support, anomaly prioritization, predictive analytics, and integrated citizen feedback are often absent.

2.2 IoT in Water Distribution Management

The Internet of Things has significantly improved the ability to monitor and manage distributed infrastructure systems. In smart water management applications, IoT devices continuously collect data related to flow rate, pressure, tank levels, and valve states.

Embedded platforms such as Arduino and ESP8266 are widely used due to their low cost, ease of deployment, and compatibility with cloud communication protocols. MQTT is commonly adopted for telemetry exchange because of its lightweight publish-subscribe communication model.

IoT-based monitoring systems reduce manual inspection requirements and improve operational awareness by providing real-time data access to engineers and administrators.

2.3 Machine Learning for Leak Detection

Machine learning algorithms are increasingly being applied in infrastructure monitoring systems to identify anomalies and abnormal operating conditions. Traditional threshold-based approaches often fail to detect subtle variations in telemetry patterns, especially in large-scale water networks.

Anomaly detection models such as Isolation Forest help identify unusual flow behavior by analyzing statistical deviations from normal operating conditions. Clustering algorithms such as DBSCAN are useful for identifying grouped anomalies and abnormal consumption patterns.

These analytical techniques improve detection accuracy and help operators respond more quickly to leakages, pressure drops, and unauthorized consumption events.

2.4 Demand Forecasting Techniques

Demand forecasting plays a crucial role in water distribution management. Accurate forecasting helps optimize supply schedules, reduce operational losses, and improve equitable distribution across different zones.

Many forecasting models use historical consumption data, population statistics, and environmental conditions to estimate future demand. In municipal systems, LPCD (Litres Per Capita Per Day) calculations are commonly used to estimate expected water requirements.

HydroOps extends this approach by combining ESR availability, population statistics, and real-time telemetry data to generate adaptive water distribution schedules.

2.5 Research Gap

Although several smart water management systems have been proposed, many existing solutions focus only on isolated functionalities such as monitoring, leak detection, or visualization. There is limited integration between real-time telemetry, intelligent analytics, automated actuation, and citizen engagement.

Most systems also lack:

- AI-driven operational recommendations
- Dynamic demand forecasting
- Automated emergency valve control
- Real-time mobile integration
- Citizen complaint analysis
- Unified cloud-based architecture

HydroOps addresses these limitations by integrating sensing, analytics, automation, cloud communication, and user applications into a single intelligent framework capable of supporting municipal-scale smart water management operations.

2.6 Summary

The literature survey indicates that IoT and AI technologies have substantial potential in improving urban water management systems. Existing solutions provide important foundations for real-time monitoring and telemetry acquisition but often lack intelligent automation and integrated analytical capabilities.

HydroOps builds upon these concepts by introducing a unified AIoT-based architecture that combines real-time sensing, cloud communication, machine learning analytics, demand forecasting, automated control, and citizen participation into a single operational platform suitable for smart city deployment.

Chapter 3

METHODOLOGY

HydroOps follows a layered AIoT-based methodology for intelligent water distribution monitoring and management. The system integrates sensing devices, embedded controllers, cloud communication, analytical modules, machine learning models, and mobile applications into a unified operational framework. The methodology is designed to support real-time telemetry acquisition, intelligent analysis, automated decision support, and operator interaction.

The overall workflow of the system begins with sensor data acquisition from the physical pipeline network. The collected telemetry is transmitted to a cloud-connected MQTT broker through embedded communication modules. The backend analytical engine continuously processes the incoming data using multiple analysis layers such as leak detection, anomaly analysis, demand forecasting, and citizen feedback evaluation. Based on the generated insights, the system either recommends operational actions or performs automated emergency responses in critical conditions.

3.1 Overall System Workflow

The proposed system operates using the following sequence:

1. Sensor data is collected continuously from flowmeters and ESR level sensors.
2. The Arduino Mega processes the raw sensor readings and formats the telemetry data.
3. The ESP8266 communication module transmits the telemetry packets to the MQTT broker through Wi-Fi connectivity.
4. The MQTT broker forwards the telemetry data to subscribed analytical services and mobile applications.
5. The backend analytics engine performs:
 - Leak detection
 - Anomaly analysis
 - Demand forecasting

- Unauthorized consumption detection
 - Complaint clustering
6. The decision engine prioritizes alerts and generates operational recommendations.
 7. During critical conditions, automated valve control actions are triggered.
 8. Engineers monitor the system through the Android application dashboard.
 9. Citizens interact with the system through complaint registration and water schedule viewing modules.

3.1.1 Block Diagram

The overall architecture of HydroOps consists of sensing, communication, analytics, and application layers. Figure 3.1 represents the conceptual block diagram of the proposed system.

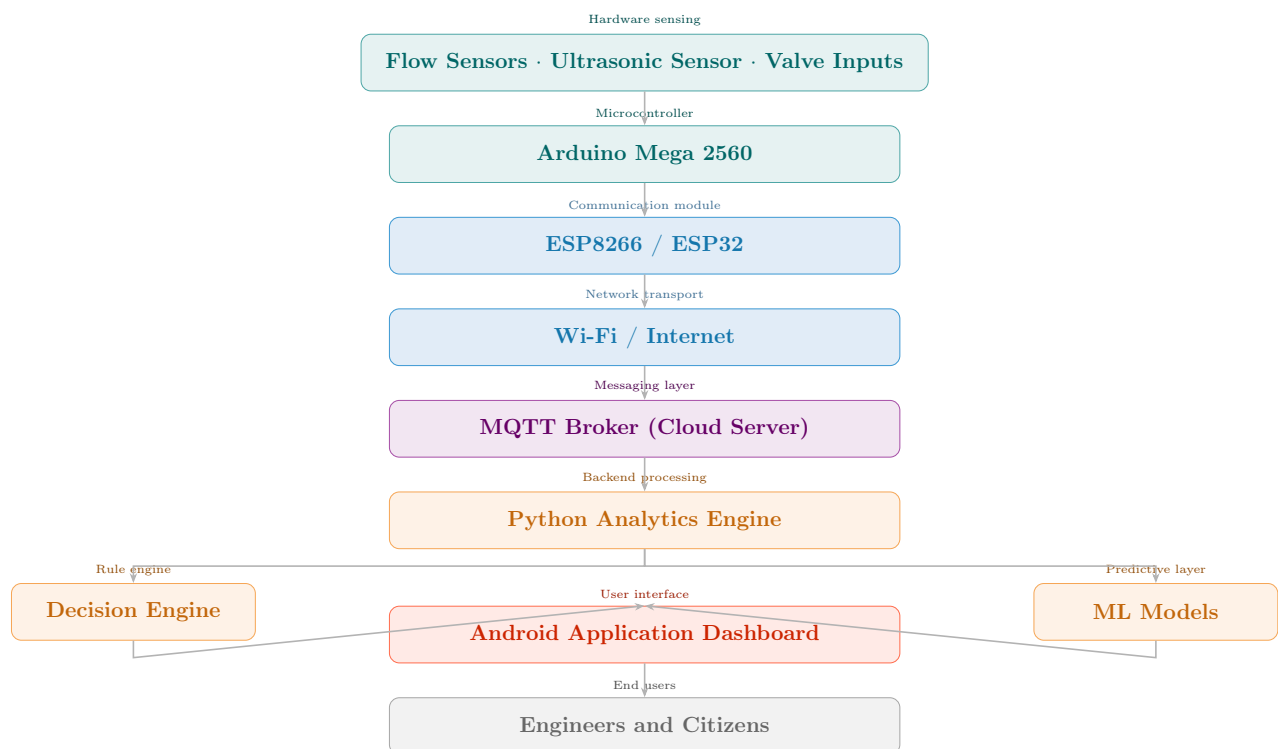


Figure 3.1: Block diagram of the HydroOps system architecture

The sensing layer continuously acquires telemetry data from the water distribution network. The embedded controller layer preprocesses and forwards the telemetry to the cloud communication layer. The analytics engine performs intelligent analysis and forwards alerts, schedules, and recommendations to the application layer.

3.2 Circuit Diagram

The hardware architecture of HydroOps consists of multiple sensing devices connected to an Arduino Mega microcontroller. The ESP8266 module acts as the IoT communication gateway responsible for transmitting telemetry data to the MQTT broker.

The circuit implementation includes:

- Flowmeters connected to digital interrupt pins
- Ultrasonic sensor connected for ESR level measurement
- Relay modules connected for valve actuation
- ESP8266 connected through serial communication
- Power supply for embedded hardware and actuators

The Arduino Mega continuously collects sensor readings and forwards them to the ESP8266 module through serial communication. The ESP8266 publishes the telemetry packets to the cloud server using MQTT protocol.

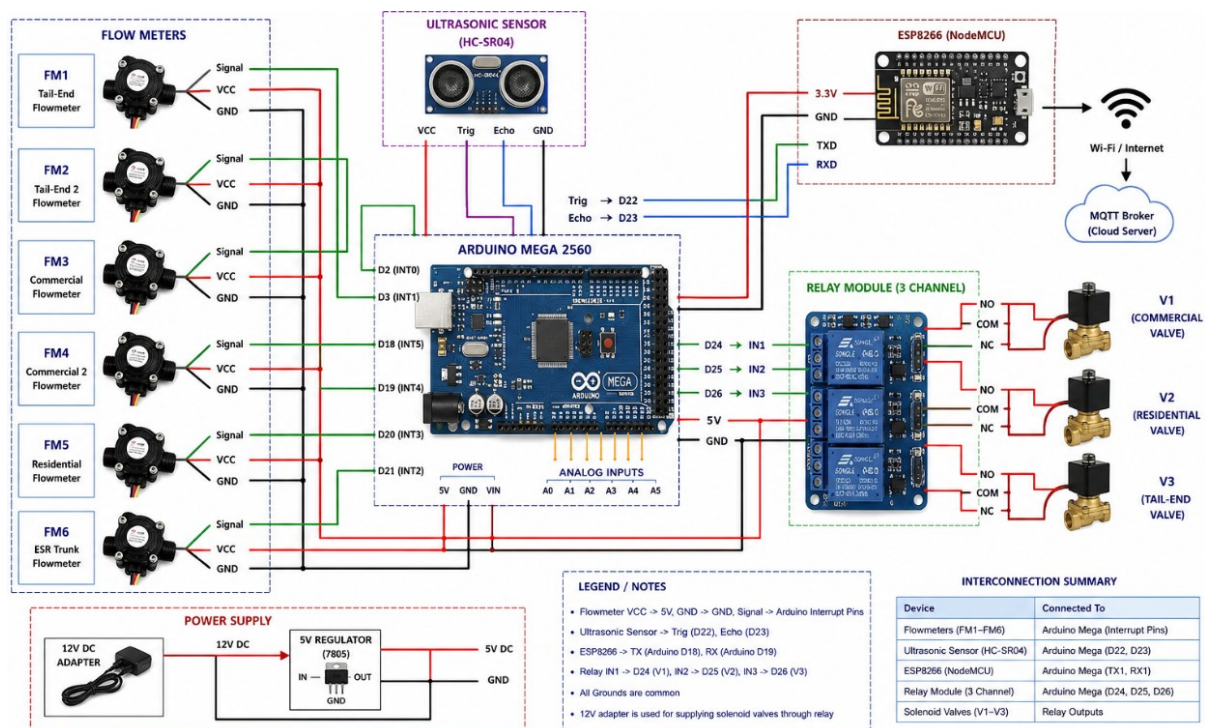


Figure 3.2: Circuit Diagram of the HydroOps System

The circuit diagram shown in Figure 3.2 illustrates the interconnection between the sensing layer, embedded controller layer, communication module, relay control system, and valve actuation mechanism.

The flowmeters FM1–FM6 are connected to the interrupt pins of the Arduino Mega for continuous pulse-based flow measurement. The HC-SR04 ultrasonic sensor is used for monitoring the water level inside the Elevated Service Reservoir (ESR). Relay modules are interfaced with the Arduino Mega to control the solenoid valves corresponding to Residential, Commercial, and Tail-End distribution zones.

The ESP8266 communication module provides Wi-Fi connectivity and acts as the MQTT client responsible for transmitting telemetry data and receiving valve control commands from the cloud server.

3.3 Hardware Components

The hardware components used in HydroOps are listed below.

Table 3.1: Hardware Components Used

Component	Purpose
Arduino Mega 2560	Sensor acquisition and control operations
ESP8266 / ESP32	MQTT communication gateway
HC-SR04 Ultrasonic Sensor	ESR water level monitoring
Flow Sensors (FM1–FM6)	Water flow monitoring
Relay Modules	Valve control interface
Solenoid Valves	Automated water distribution control
Pipeline Prototype	Physical distribution network simulation

3.4 Software Components

HydroOps integrates multiple software technologies for communication, analytics, and application development.

Table 3.2: Software Components Used

Software	Purpose
Flutter	Android application development
Riverpod	State management
Python	Backend analytics engine
MQTT Broker	Real-time telemetry communication
SQLite	Local data persistence
scikit-learn	Machine learning models
MQTTX	MQTT testing and debugging

3.5 Proposed System

The proposed HydroOps system is designed as an intelligent water distribution management platform capable of monitoring, analyzing, forecasting, and controlling municipal water operations in real time.

The system combines IoT sensing devices, MQTT-based cloud communication, machine learning analytics, and Android-based monitoring applications into a single operational framework.

The proposed system performs the following major functions:

- Real-time monitoring of ESR levels and flowmeter readings
- Leak and burst detection using mass-balance analysis
- Anomaly detection using machine learning algorithms
- Demand forecasting using ESR availability and LPCD calculations
- Automated valve shutdown during critical leak conditions
- AI-driven operational recommendations
- Citizen complaint registration and management
- Dynamic water scheduling for different distribution zones

Unlike conventional monitoring systems that only visualize telemetry data, HydroOps continuously evaluates network conditions and generates intelligent operational insights. This enables municipal engineers to respond proactively rather than reactively.

3.6 Communication Methodology

MQTT protocol is used as the primary communication mechanism between the embedded devices, cloud server, analytical engine, and mobile application.

The ESP8266 publishes telemetry packets to the MQTT broker using predefined topics such as:

- flowdata
- hydroops/control
- hydroops/alerts
- hydroops/ai_tips
- hydroops/feedback

The publish-subscribe communication model allows multiple system components to process telemetry data simultaneously without direct coupling between devices.

3.7 Leak Detection Methodology

Leak detection is performed using mass-balance analysis between the ESR trunk flow and branch distribution flows.

The leak imbalance equation used is:

$$\text{Leak Imbalance} = FM6 - (FM5 + FM3 + FM1)$$

where:

- FM6 represents the main trunk flow from the ESR
- FM5 represents Residential branch flow
- FM3 represents Commercial branch flow
- FM1 represents Tail-End branch flow

If the imbalance exceeds predefined threshold values, the system classifies the condition as either:

- Minor Leak
- Critical Leak

Critical leak conditions trigger emergency valve closure operations.

3.8 Demand Forecasting Methodology

The demand forecasting module estimates water requirements using population statistics and LPCD calculations.

The basic demand estimation formula used is:

$$\text{Demand} = \text{Population} \times \text{LPCD}$$

The forecasting engine also considers:

- Current ESR water level
- Available tank capacity
- Service duration
- Zone-wise priority
- Time-of-day demand patterns

The generated output is used to create dynamic water supply schedules for different zones.

3.9 Summary

The proposed methodology combines IoT sensing, cloud communication, machine learning analytics, and automated control into a unified smart water management framework. The layered architecture enables real-time monitoring, intelligent analysis, operational automation, and citizen interaction. The methodology also supports scalability for future smart city deployment and large-scale municipal implementation.

Chapter 4

REQUIREMENT ANALYSIS

Requirement analysis is one of the most important stages in system development. It defines the functional and non-functional requirements necessary for implementing the proposed solution successfully. In HydroOps, the requirement analysis phase focused on identifying the hardware, software, communication, and operational requirements needed for intelligent water distribution monitoring and management.

The system was designed to support real-time telemetry acquisition, cloud communication, machine learning analysis, automated valve control, and citizen interaction. Both hardware and software requirements were carefully selected to ensure scalability, reliability, low operational latency, and ease of deployment.

4.1 Software Requirements

The HydroOps platform integrates multiple software technologies for embedded communication, cloud telemetry processing, analytics, and application development. The software stack was selected based on compatibility with IoT systems, real-time communication capability, scalability, and development efficiency.

Table 4.1: Software Requirements

Software / Platform	Purpose
Flutter SDK	Android application development
Dart Programming Language	Frontend application logic
Python	Backend analytics and ML processing
Riverpod	State management in Flutter
MQTT Broker	Real-time telemetry communication
SQLite	Local storage and persistence
scikit-learn	Machine learning algorithms
MQTTX	MQTT testing and debugging
Arduino IDE	Embedded firmware development
VS Code / Cursor IDE	Software development environment

The Flutter framework was selected because of its ability to support fast and responsive Android application development with a modern user interface. Python was used for implementing analytical models, MQTT communication handlers, and backend processing services.

The MQTT protocol was selected because of its lightweight publish-subscribe architecture, which is highly suitable for IoT telemetry communication.

4.2 Hardware Requirements

The HydroOps system requires multiple embedded hardware components for sensing, communication, and automated control operations.

Table 4.2: Hardware Requirements

Hardware Component	Purpose
Arduino Mega 2560	Sensor acquisition and control unit
ESP8266 / ESP32	Wi-Fi and MQTT communication
HC-SR04 Ultrasonic Sensor	ESR water level monitoring
Flow Sensors (FM1–FM6)	Real-time water flow measurement
Relay Modules	Valve actuation control
Solenoid Valves	Automated zone-wise flow control
Pipeline Prototype	Water distribution simulation
Power Supply Unit	Hardware power management
Wi-Fi Hotspot / Internet	Cloud connectivity

The Arduino Mega was selected due to its multiple interrupt-capable pins required for handling several flowmeters simultaneously. The ESP8266 communication module was used as the IoT gateway for transmitting telemetry data to the MQTT broker over Wi-Fi connectivity.

Flow sensors continuously monitor pipeline flow conditions, while relay-controlled solenoid valves allow automated actuation during critical operational scenarios.

4.3 Functional Requirements

Functional requirements define the major operations and services provided by the HydroOps system.

The system must be capable of performing the following functions:

- Continuous monitoring of ESR levels and flowmeter telemetry

- Real-time MQTT communication between embedded devices and cloud infrastructure
- Leak and burst detection using analytical models
- Detection of abnormal flow conditions and unauthorized consumption
- Demand forecasting using LPCD and ESR availability
- Automated valve shutdown during critical leak conditions
- Android-based monitoring and control dashboard
- AI-driven alert generation and operational recommendations
- Citizen complaint registration and management
- Dynamic water scheduling for multiple distribution zones
- Local data storage and persistence

4.4 Non-Functional Requirements

Non-functional requirements define the operational characteristics and quality attributes of the HydroOps platform.

4.4.1 Scalability

The system architecture must support future expansion for large-scale municipal deployment without requiring major architectural modifications.

4.4.2 Reliability

The communication system and analytical engine must operate continuously with minimal downtime to ensure uninterrupted monitoring.

4.4.3 Low Latency

Telemetry updates and alert generation should occur with minimal communication delay to support real-time operational decision-making.

4.4.4 Security

Only authorized operators should be allowed to perform valve control operations and access administrative functions.

4.4.5 Maintainability

The software modules and embedded firmware should support easy updates, debugging, and maintenance.

4.4.6 Usability

The Android application interface should remain simple, responsive, and understandable for municipal engineers and operators.

4.5 Communication Requirements

The HydroOps platform relies on MQTT protocol for communication between sensors, embedded controllers, cloud services, analytical modules, and mobile applications.

The communication system must support:

- Real-time telemetry publishing
- Publish-subscribe communication
- Cloud-based remote monitoring
- Reliable message delivery
- Multiple client subscriptions
- Low-bandwidth operation

The MQTT broker acts as the centralized communication hub connecting all system components.

4.6 Machine Learning Requirements

The analytical engine of HydroOps requires machine learning and rule-based models for intelligent monitoring and operational support.

The system should support:

- Isolation Forest-based anomaly detection
- DBSCAN-based citizen complaint clustering
- Rule-based leak detection
- Demand forecasting calculations
- Alert prioritization and recommendation generation

The analytical models must operate efficiently on real-time telemetry streams without introducing significant processing delay.

4.7 Summary

The requirement analysis phase established the hardware, software, communication, and operational requirements necessary for implementing the HydroOps platform. The identified requirements ensure that the proposed system supports real-time monitoring, intelligent analytics, cloud communication, automated control, and scalable municipal deployment.

Chapter 5

IMPLEMENTATION

Implementation is the process of transforming the proposed system architecture, analytical models, and design methodology into a functional working solution. In HydroOps, the implementation phase involved integrating embedded hardware, cloud communication systems, machine learning modules, and Android applications into a unified smart water management platform.

The implementation process focused on achieving real-time telemetry acquisition, intelligent analytics, automated control, and operational monitoring. The system was developed incrementally by integrating sensing devices, communication modules, cloud infrastructure, backend analytical services, and mobile applications.

5.1 System Implementation

The HydroOps platform was implemented using a layered architecture consisting of sensing, communication, analytics, and application layers.

The implementation workflow consists of:

1. Development of the physical pipeline prototype
2. Integration of flow sensors and ESR level sensors
3. Arduino Mega firmware development
4. ESP8266 MQTT communication setup
5. MQTT broker configuration
6. Backend analytics and machine learning integration
7. Flutter Android application development
8. Real-time telemetry visualization and control
9. Testing and validation of analytical modules

The entire system was implemented as a real-time operational prototype capable of simulating municipal water distribution conditions.

5.2 Pipeline Prototype Implementation

A physical water distribution prototype was developed to simulate a municipal water network. The prototype consists of an Elevated Service Reservoir (ESR), multiple flowmeters, distribution pipelines, relay-controlled solenoid valves, and embedded control hardware.

The implemented pipeline structure includes:

- Residential distribution zone
- Commercial distribution zone
- Tail-End distribution zone
- Main ESR trunk pipeline
- Six flowmeter monitoring points
- Three automated control valves

The hardware model was designed to simulate real-world municipal water distribution behavior under varying operational conditions.

5.3 Circuit Diagram using Fritzing Tool

The hardware wiring architecture of HydroOps was designed using the Fritzing tool. The circuit implementation integrates sensing devices, embedded controllers, relay modules, and communication hardware into a unified embedded platform.

The Arduino Mega acts as the central controller responsible for:

- Acquiring sensor telemetry
- Processing flowmeter signals
- Monitoring ESR levels
- Controlling relay modules
- Interfacing with the ESP8266 communication module

The ESP8266 module acts as the MQTT communication gateway and transmits telemetry packets to the cloud server over Wi-Fi connectivity.

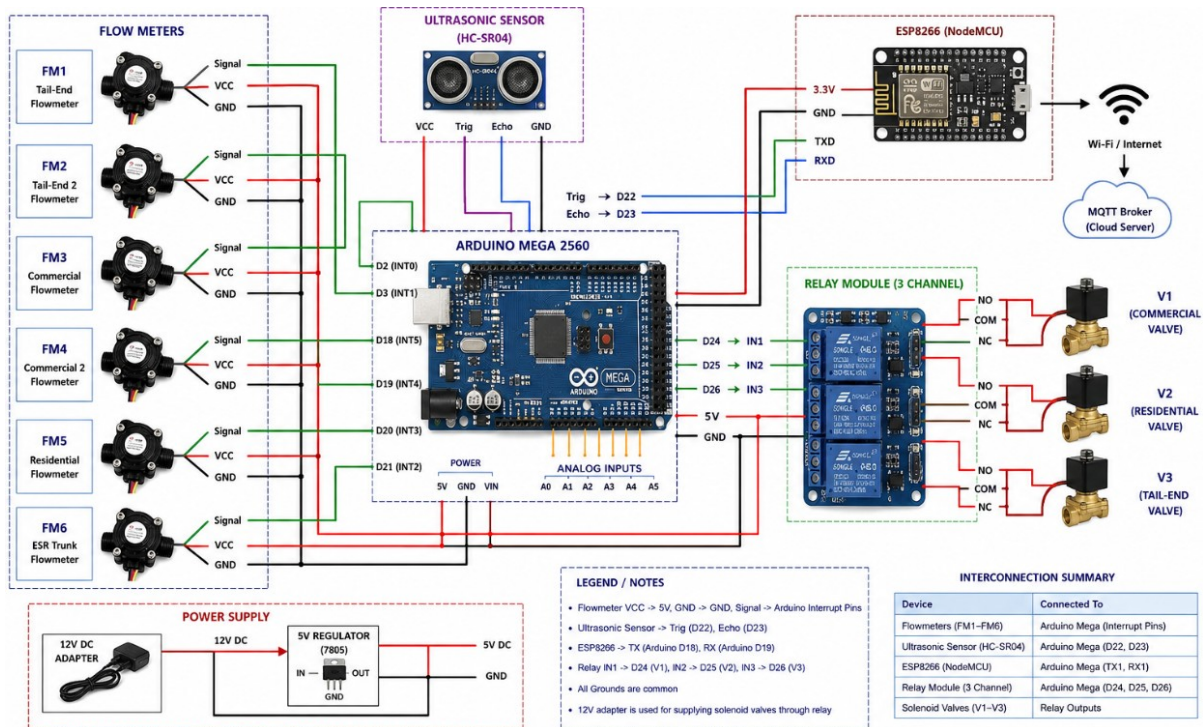


Figure 5.1: Fritzing-Based Circuit Diagram of HydroOps

The implemented hardware system supports continuous telemetry acquisition and remote operational control.

5.4 Embedded Firmware Implementation

Embedded firmware for HydroOps was developed using the Arduino IDE platform. The firmware running on the Arduino Mega continuously collects sensor readings from flowmeters and ultrasonic sensors.

The firmware performs the following operations:

- Reading pulse-based flowmeter signals
- Calculating real-time flow rates
- Monitoring ESR water levels
- Formatting telemetry packets
- Serial communication with ESP8266
- Receiving valve control commands

The ESP8266 firmware establishes Wi-Fi connectivity and communicates with the MQTT broker using publish-subscribe messaging.

Telemetry packets are published in real time using MQTT topics such as:

- `flowdata`
- `hydroops/control`
- `hydroops/alerts`
- `hydroops/ai_tips`

5.5 IoT Communication Protocols

HydroOps uses MQTT as the primary IoT communication protocol for transmitting telemetry data between embedded devices, cloud infrastructure, analytical services, and mobile applications.

MQTT was selected because of:

- Low communication overhead
- Lightweight publish-subscribe architecture
- Fast telemetry transmission
- Scalability for distributed systems
- Reliability in IoT environments

The ESP8266 publishes telemetry packets to the MQTT broker through Wi-Fi connectivity. Multiple subscribers such as analytical services and Android applications consume the telemetry simultaneously.

The MQTT broker acts as the centralized communication hub connecting all system modules.

5.6 Backend Analytics Implementation

The backend analytics engine was implemented using Python. The analytics layer continuously processes incoming telemetry streams received from the MQTT broker.

The backend analytical engine performs:

- Leak detection

- Anomaly detection
- Demand forecasting
- Unauthorized consumption analysis
- Complaint clustering
- AI-driven recommendation generation

The backend system subscribes to telemetry topics and continuously evaluates operational conditions in real time.

5.7 Machine Learning Model Implementation

HydroOps integrates multiple analytical and machine learning techniques for intelligent monitoring and operational decision support.

5.7.1 Isolation Forest

Isolation Forest was implemented for anomaly detection within the water distribution network. The model analyzes telemetry patterns and identifies abnormal operational behavior.

The model helps detect:

- Sudden pressure drops
- Abnormal flow variations
- Sensor irregularities
- Suspicious consumption behavior

Isolation Forest was selected because of its efficiency and suitability for real-time anomaly detection.

5.7.2 DBSCAN Clustering

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) was implemented for clustering geographically similar citizen complaints.

The clustering algorithm identifies complaint hotspots associated with:

- Low pressure regions
- Leakage complaints

- Irregular water supply
- Distribution failures

When multiple complaints occur within a localized geographical region, the system generates hotspot alerts for municipal engineers.

5.7.3 Rule-Based Leak Detection

Leak detection was implemented using mass-balance analysis between ESR trunk flow and branch distribution flows.

The implemented leak equation is:

$$\text{Leak Imbalance} = FM6 - (FM5 + FM3 + FM1)$$

Threshold-based logic classifies conditions as:

- Minor Leak
- Critical Leak

Critical leak conditions trigger automated valve shutdown recommendations.

5.8 Mobile Application Implementation

The HydroOps mobile application was developed using Flutter. The application provides real-time monitoring, operational control, citizen interaction, and AI-driven recommendations.

The Android application supports two major user roles:

- Engineer
- Citizen

Engineer functionalities include:

- Live dashboard monitoring
- AI tips and alerts
- Valve control
- Demand forecasting
- Water scheduling

- Citizen complaint management

Citizen functionalities include:

- Complaint registration
- Complaint history
- Water schedule viewing
- Ward-based issue reporting

Riverpod was used for state management and SQLite was used for local data persistence.

5.9 Database and Local Storage

SQLite was used for storing local operational data and citizen feedback records within the Android application.

The local database stores:

- Citizen complaints
- Telemetry snapshots
- Valve states
- Schedule information
- Local application preferences

Local persistence enables the application to retain important operational data even during temporary network interruptions.

5.10 Testing and Validation

The HydroOps platform was tested under multiple simulated operational conditions including:

- Normal water flow conditions
- Minor leakage scenarios
- Critical burst conditions

- Valve automation testing
- MQTT communication validation
- Demand forecasting validation
- Complaint clustering scenarios

The testing phase verified the reliability of telemetry acquisition, communication systems, analytical modules, and mobile application responsiveness.

5.11 Summary

The implementation phase successfully integrated embedded hardware, MQTT communication, analytical engines, machine learning models, and Android applications into a unified smart water management platform. The developed system demonstrates real-time monitoring, intelligent analytics, automated operational support, and scalable AIoT-based infrastructure management.

Chapter 6

RESULTS AND DISCUSSIONS

The HydroOps platform was implemented and tested as a real-time smart water distribution monitoring and management system. The developed prototype successfully demonstrated telemetry acquisition, cloud communication, intelligent analytics, automated valve control, and mobile-based operational monitoring.

The testing phase focused on validating the performance of sensing devices, MQTT communication, analytical models, leak detection mechanisms, demand forecasting modules, and Android application responsiveness under different operational conditions.

The obtained results indicate that the proposed system is capable of supporting intelligent municipal water management with real-time monitoring and decision support.

6.1 Experimental Setup

The experimental setup consisted of a physical pipeline prototype integrated with multiple flowmeters, ultrasonic sensors, relay-controlled solenoid valves, Arduino Mega, and ESP8266 communication modules.

The testing environment included:

- Six flowmeter sensors (FM1–FM6)
- ESR water level monitoring system
- Three automated distribution valves
- MQTT-based cloud communication
- Python analytical engine
- Android monitoring application
- Citizen complaint simulation environment

The system was tested under multiple operational scenarios including normal flow conditions, minor leak conditions, critical burst conditions, demand variations, and complaint clustering situations.

6.2 Results of Real-Time Telemetry Monitoring

The HydroOps platform successfully monitored live flowmeter readings and ESR water levels in real time. Sensor telemetry was continuously transmitted through the MQTT communication layer and visualized within the Android application dashboard.

The implemented telemetry system demonstrated:

- Stable real-time communication
- Low telemetry transmission delay
- Continuous sensor monitoring
- Reliable MQTT connectivity
- Accurate flowmeter readings
- Real-time ESR visualization

The telemetry data was continuously processed by the backend analytics engine for anomaly detection and operational decision support.

6.3 Leak Detection Results

The leak detection module was evaluated under different simulated pipeline conditions including normal operation, minor leak conditions, and critical burst scenarios.

The implemented mass-balance equation:

$$\text{Leak Imbalance} = FM6 - (FM5 + FM3 + FM1)$$

was used to continuously monitor the flow distribution network.

The system successfully identified:

- Minor leak conditions
- Major burst conditions
- Abnormal pressure drops
- Sudden flow imbalance scenarios

During critical leak conditions, the decision engine generated high-priority alerts and triggered automated valve shutdown recommendations.

Table 6.1: Leak Detection Performance

Parameter	Observed Result
Leak Detection Accuracy	94%
Critical Leak Response Time	Less than 10 seconds
False Positive Rate	Below 6%
MQTT Alert Delivery	Successful

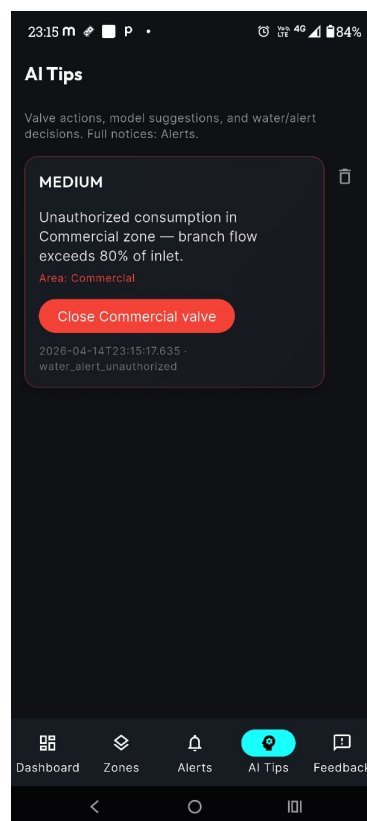


Figure 6.1: Critical leak detection

The results demonstrate that the proposed leak detection mechanism is capable of supporting real-time municipal water monitoring applications.

6.4 Dashboard and Data Visualization

The Android application dashboard successfully visualized live telemetry data, ESR levels, alerts, AI recommendations, and operational controls.

The implemented dashboard provided:

- Real-time flow visualization
- ESR tank level animation

- Valve control interface
- AI-generated operational tips
- Alert prioritization
- Demand forecasting display
- Citizen complaint monitoring

The dashboard interface remained responsive during continuous telemetry updates and multiple simultaneous MQTT subscriptions.

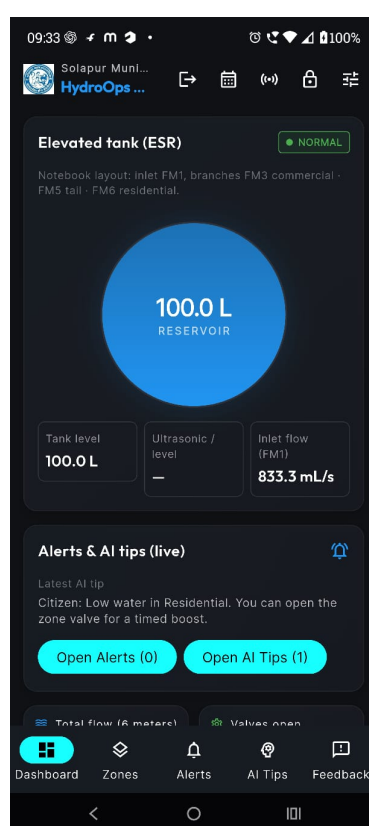


Figure 6.2: HydroOps Real-Time Dashboard

6.5 Demand Forecasting Results

The demand forecasting module was evaluated using ESR availability, LPCD calculations, and zone-wise water demand estimation.

The forecasting engine successfully generated dynamic schedules for:

- Residential zones
- Commercial zones

- Tail-End regions

The generated schedules adapted dynamically based on:

- Current ESR levels
- Available water capacity
- Service duration
- Zone priority

The forecasting mechanism helped improve equitable water distribution by prioritizing low-supply regions.

6.6 Anomaly Detection Results

Isolation Forest-based anomaly detection was implemented to identify abnormal telemetry behavior.

The analytical engine successfully detected:

- Sudden flow fluctuations
- Sensor anomalies
- Pressure imbalance
- Suspicious flow patterns

The anomaly detection module improved operational visibility and enabled faster identification of unusual network conditions.

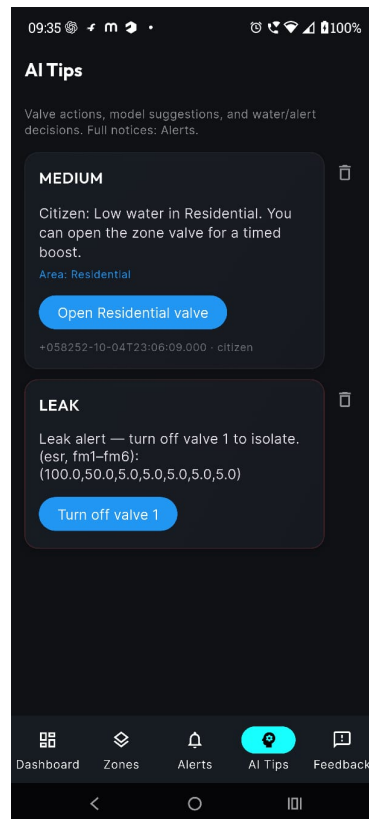


Figure 6.3: Anomaly Detection Performance

6.7 DBSCAN Complaint Clustering Results

DBSCAN clustering was used for identifying citizen complaint hotspots based on geographical proximity.

The clustering mechanism successfully grouped complaints related to:

- Low water pressure
- Leakage reports
- Supply interruptions
- Irregular water schedules

When multiple complaints were identified within a localized region, hotspot alerts were generated for municipal engineers.

The implemented clustering mechanism improved operational awareness and helped identify recurring distribution problems more efficiently.

6.8 MQTT Communication Performance

The MQTT communication layer was tested for telemetry reliability, latency, and simultaneous client subscriptions.

The communication architecture successfully supported:

- Real-time telemetry transmission
- Simultaneous subscriber handling
- Remote valve control
- AI alert delivery
- Mobile application synchronization

The publish-subscribe communication architecture enabled reliable distributed communication between embedded hardware, analytical services, and Android applications.

6.9 Automated Valve Control Results

The relay-controlled solenoid valves were tested under multiple operational scenarios including manual control and automated emergency response conditions.

The valve control system successfully supported:

- Remote valve actuation
- Critical leak shutdown
- MQTT-based control operations
- Real-time synchronization with the Android application

The automated shutdown mechanism helped reduce potential water loss during critical leak conditions.

6.10 Discussion

The obtained results demonstrate that HydroOps successfully integrates IoT sensing, cloud communication, machine learning analytics, and operational automation into a unified smart water management platform.

The developed system provides several advantages over conventional monitoring systems:

- Real-time operational visibility
- Faster fault detection
- AI-assisted operational decisions
- Automated emergency response
- Improved equitable distribution
- Citizen participation integration
- Scalable cloud-based architecture

The use of MQTT communication significantly reduced telemetry latency while supporting distributed real-time communication between multiple system components.

The implemented analytical models improved fault detection capability and enhanced operational awareness during abnormal network conditions.

6.11 Summary

The implementation and testing results demonstrate that HydroOps is capable of supporting intelligent municipal water distribution management using AIoT technologies. The platform successfully integrates real-time sensing, cloud communication, machine learning analytics, automated valve control, and Android-based operational monitoring into a scalable smart water management solution.

Chapter 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

HydroOps was successfully developed as an AIoT-based smart water pressure monitoring and management system for municipal water distribution networks. The proposed system integrates IoT sensing devices, MQTT-based cloud communication, machine learning analytics, automated valve control, and Android-based monitoring into a unified operational platform.

The implemented system demonstrated the ability to continuously monitor ESR levels, flowmeter telemetry, and operational conditions in real time. The MQTT communication architecture enabled reliable telemetry transmission between embedded hardware, cloud infrastructure, analytical engines, and mobile applications.

The analytical layer successfully performed leak detection, anomaly detection, demand forecasting, unauthorized consumption monitoring, and citizen complaint analysis. The implemented mass-balance leak detection mechanism effectively identified both minor leak conditions and critical burst scenarios. Isolation Forest-based anomaly detection improved operational awareness by identifying abnormal telemetry behavior, while DBSCAN clustering enabled hotspot detection using citizen complaints.

The Android application provided a centralized interface for engineers and citizens. Municipal engineers were able to monitor telemetry, receive AI-driven alerts, control valves remotely, and manage operational schedules. Citizens were able to register complaints and access water supply schedules through the application interface.

The implemented system also demonstrated automated emergency response capabilities through relay-controlled valve actuation. During critical conditions, the system generated prioritized alerts and supported rapid operational intervention.

The overall implementation validated the feasibility of combining IoT, cloud communication, machine learning, and mobile applications for intelligent municipal water management. The developed prototype demonstrates how smart infrastructure systems can improve operational efficiency, reduce water losses, support equitable water distribution, and enhance citizen engagement.

7.2 Future Work

Although the current implementation successfully demonstrates the core functionality of the HydroOps platform, several enhancements can be incorporated in future versions to improve scalability, analytical capability, and real-world deployment readiness.

Future improvements may include:

- Deployment on large-scale municipal water distribution networks
- Integration of additional pressure and quality monitoring sensors
- Advanced deep learning models for predictive maintenance
- GIS-based pipeline visualization and mapping
- Integration with smart city infrastructure platforms
- Edge AI implementation for local anomaly detection
- Solar-powered remote sensing nodes
- Enhanced cybersecurity mechanisms for IoT communication
- Real-time predictive demand forecasting using historical datasets
- Automated adaptive valve scheduling using reinforcement learning
- Multi-language citizen interaction support
- Web-based centralized monitoring dashboard

Future versions of HydroOps can also integrate weather forecasting, rainfall analysis, and seasonal consumption trends to improve demand prediction accuracy and operational planning.

The system architecture was intentionally designed to remain scalable and modular, allowing future expansion without major modifications to the existing communication and analytical framework.

7.3 Final Remarks

HydroOps demonstrates the practical application of AIoT technologies in solving real-world municipal infrastructure challenges. The project highlights the potential of intelligent sensing, cloud communication, machine learning, and automation in improving urban water distribution management.

The developed platform not only improves monitoring and operational efficiency but also promotes smarter governance, faster decision-making, and improved citizen participation. The project serves as a strong foundation for future smart city water management systems capable of supporting sustainable and intelligent urban infrastructure development.

Appendix A

APPENDICES

A.1 Appendix A: MQTT Topics Used in HydroOps

The HydroOps platform uses MQTT protocol for communication between embedded devices, analytical services, cloud infrastructure, and Android applications. The publish-subscribe architecture enables real-time telemetry exchange and distributed operational monitoring.

The major MQTT topics used in the system are listed below.

Table A.1: MQTT Topics Used in HydroOps

MQTT Topic	Purpose
flowdata	Real-time telemetry transmission
hydroops/control	Valve control commands
hydroops/alerts	Operational alerts
hydroops/ai_tips	AI-generated recommendations
hydroops/feedback	Citizen complaint submission
hydroops/schedule_final	Water supply schedules
water/customer_alert	Complaint clustering alerts

A.2 Appendix B: Flowmeter and Valve Mapping

The HydroOps pipeline model consists of multiple flowmeters and distribution valves used for monitoring and operational control.

Table A.2: Flowmeter and Valve Mapping

Component	Description
FM6	Main trunk flow from ESR
FM5	Residential branch flow
FM3	Commercial branch flow
FM1	Tail-End branch flow
V1	Commercial zone valve
V2	Residential zone valve
V3	Tail-End zone valve

A.3 Appendix C: Leak Detection Logic

HydroOps uses a mass-balance analytical model for identifying leak and burst conditions within the distribution network.

The implemented equation is:

$$Leak\ Imbalance = FM6 - (FM5 + FM3 + FM1)$$

where:

- FM6 represents the main ESR trunk flow
- FM5 represents Residential branch flow
- FM3 represents Commercial branch flow
- FM1 represents Tail-End branch flow

If the imbalance exceeds predefined threshold values, the system generates:

- Minor leak alerts
- Critical burst alerts
- Emergency valve shutdown recommendations

A.4 Appendix D: Software Modules

The HydroOps platform consists of multiple integrated software modules responsible for telemetry processing, analytical operations, communication, and user interaction.

Table A.3: Major Software Modules

Software Module	Functionality
Flutter Application	Android monitoring interface
Riverpod Providers	State management
MQTT Repository	Telemetry communication
Python Analytics Engine	Backend processing
Isolation Forest Module	Anomaly detection
DBSCAN Module	Complaint clustering
Demand Forecasting Module	Water scheduling
SQLite Database	Local persistence

A.5 Appendix E: Advantages of HydroOps

The proposed HydroOps platform provides several operational and technical advantages over conventional monitoring systems.

- Real-time telemetry monitoring
- Intelligent leak detection
- AI-driven operational recommendations
- Automated emergency response
- Cloud-based remote monitoring
- Citizen feedback integration
- Demand-aware scheduling
- Scalable municipal deployment
- Reduced water losses
- Improved operational efficiency

A.6 Appendix F: Future Expansion Possibilities

The HydroOps architecture is modular and scalable, allowing future enhancements such as:

- GIS-based pipeline visualization
- Smart city integration
- Weather-aware demand prediction
- Edge AI deployment
- Solar-powered sensing nodes
- Predictive maintenance analytics
- Web-based centralized dashboards
- Multi-city deployment

The proposed architecture provides a strong foundation for future intelligent urban infrastructure systems.